

MAE339 AEROSPACE ENG. LAB

Lab 2 Report

Humza Khan - 50363412



INTRODUCTION



Theory and Background

The concept of **Flow Visualization (Fig 1.)** is being focused on during this experiment. Qualitatively describing how the visual streamlines (highlighted with various buoyant dyes) change and interact with the changing **Reynold's number (Fig 2.)** (which helps us determine whether the flow is laminar or turbulent) and the changing **angle of attack (AoA)** of the experimental objects placed in the flowing liquid.

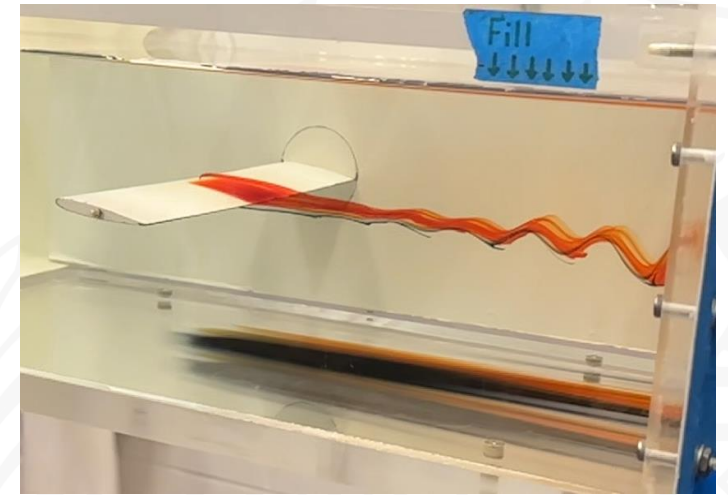


Fig 1. Flow Visualization via Streamlines

$$Re = \frac{\rho u L}{\mu}$$

Fig 2. Reynold's Number Formula
(Essentially a ratio of Inertial forces & viscous forces in a fluid)

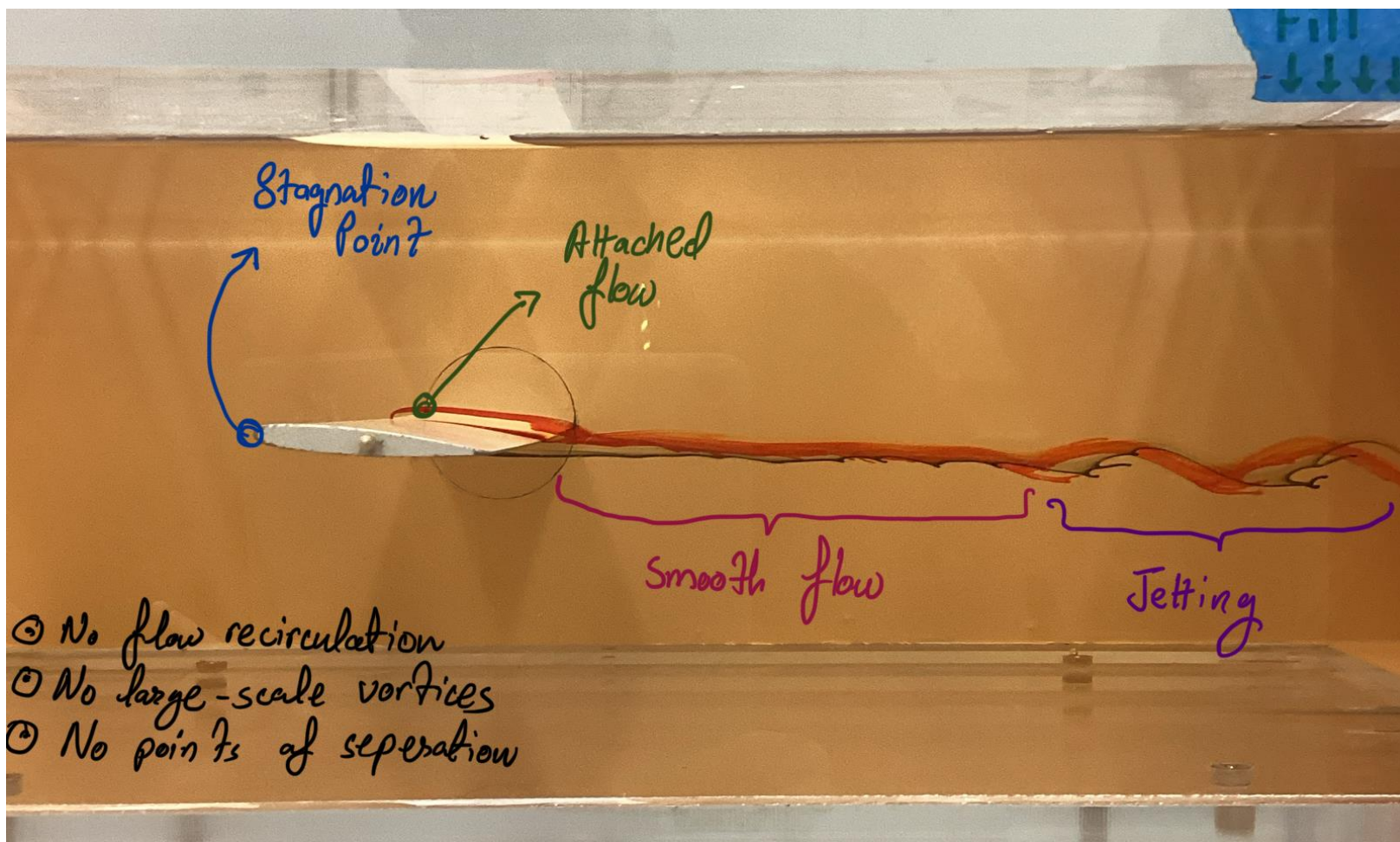
Experimental Setup

- We were provided with a closed circuit water channel which we filled up until the 'fill line' using the filling valve. This was done after the drain valve was closed, and pump valve was ensured that it was open.
- Team 2 made the neutrally buoyant dyes for the experiment. They tested and made 2 dyes which were orange and purple in color.
- I was on Team 1, therefore we installed different experimental solids in the closed circuit water channel. My team was also responsible for washing and bleeding the system of any previous remnant dyes and air bubbles
- Once everything was set up according to the manual, both teams combined for data collection, which included tasks like note-taking, imaging, and recording videos. The members not involved with data collection operated the water pump and made required changes to apparatus as the experiment progressed forward.

AIRFOIL RESULTS & DISCUSSION

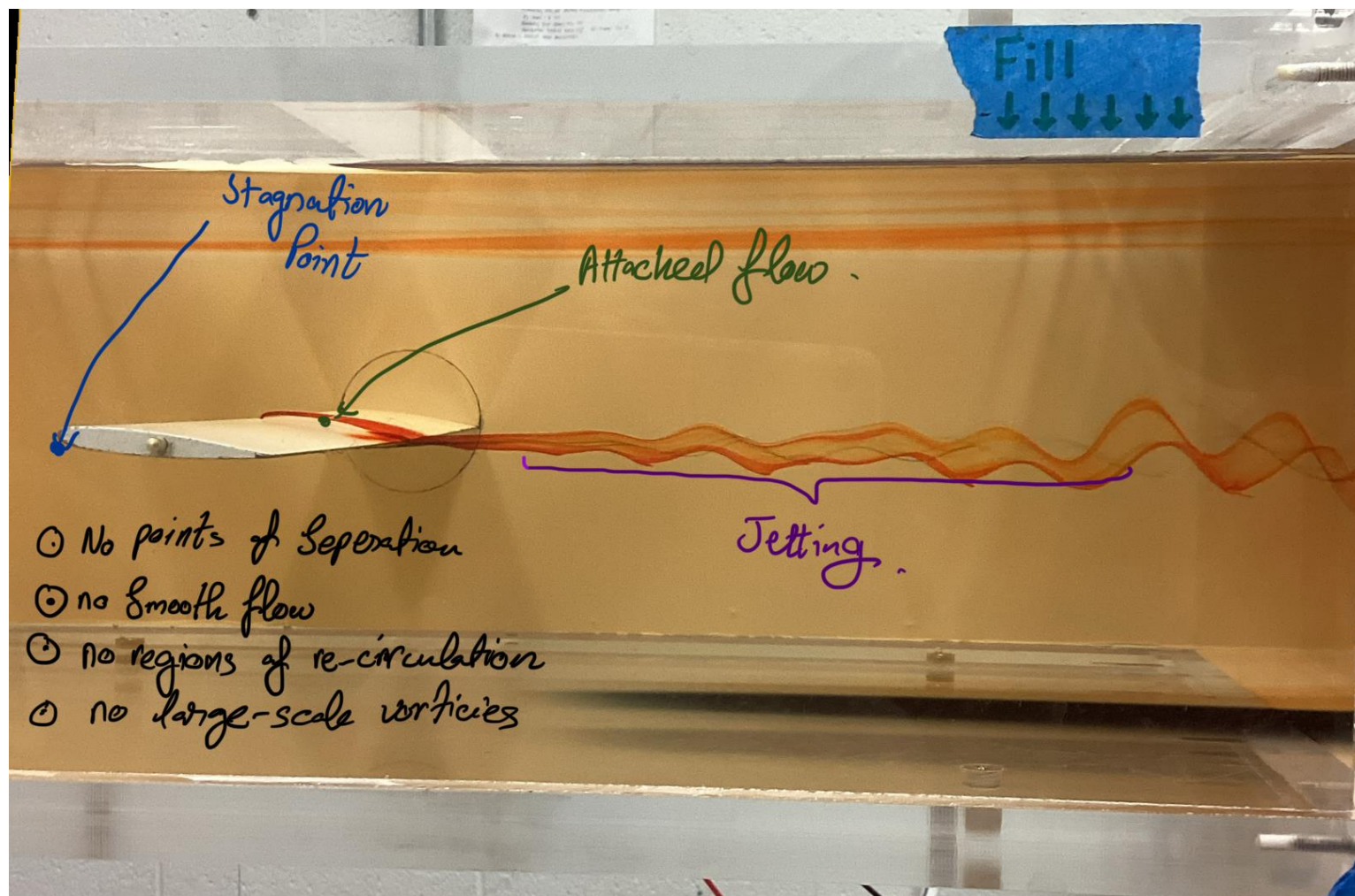


Airfoil - Result (1): Low Reynold's Number and AoA



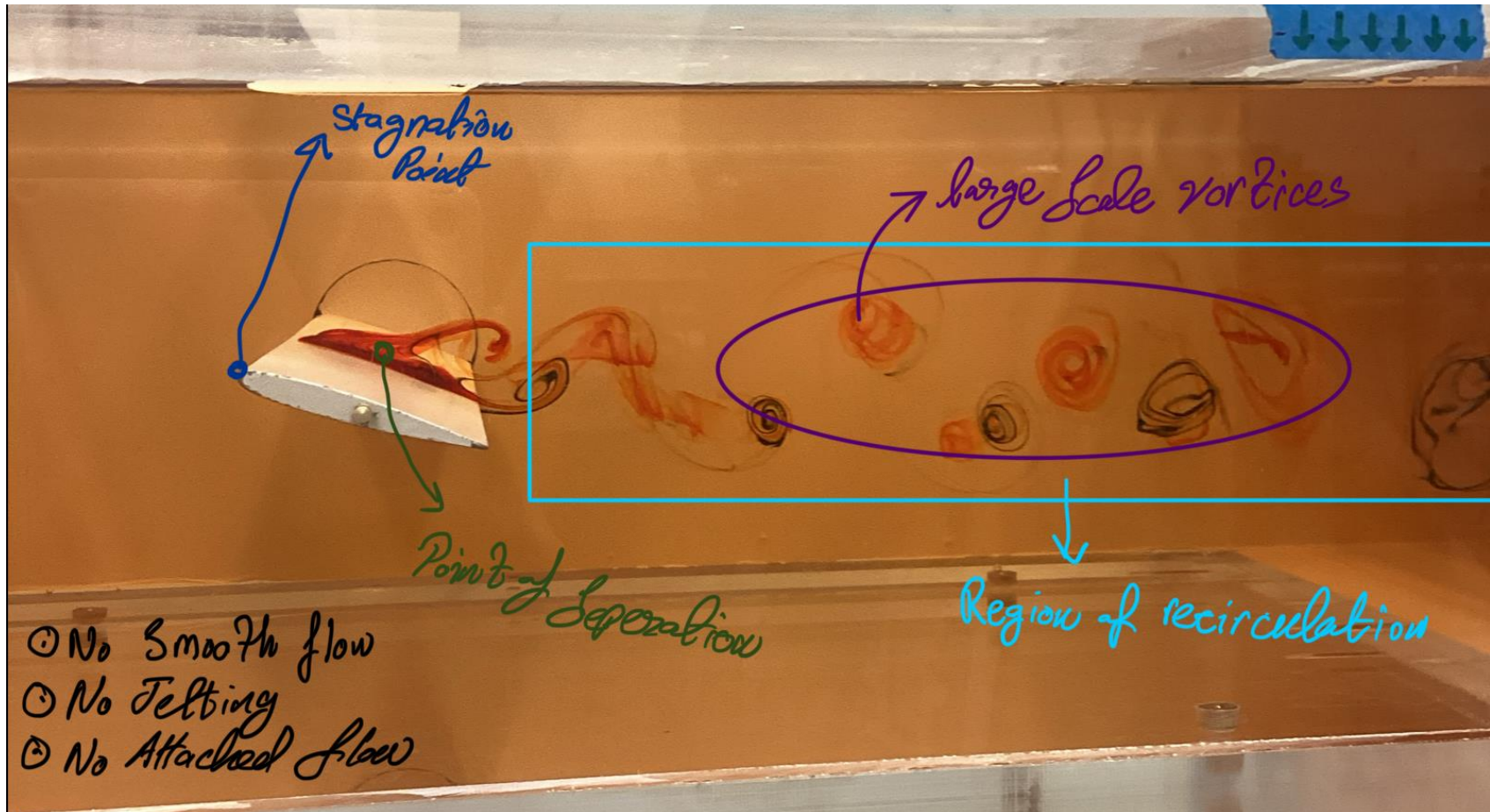
- Estimated Reynold's Number: **~800**
- Measured AoA: **~4°**

Airfoil - Result (2): Higher Reynold's Number and Low AoA



- Estimated Reynold's Number: **~1200**
- Measured AoA: **~4°**

Airfoil - Result (3): Low Reynold's Number and High AoA



- Estimated Reynold's Number: **~800**
- Measured AoA: **~14°**

Airfoil - Discussion

1. At lower AoA and Reynold's number, the system seems to be in a very stable and steady. The flow is evenly attached to the body and re-joins steadily at the trailing edge of the airfoil, followed by smooth flow and a little jetting.

As the Reynold's number increases, although the flow remains steady and attached to the airfoil, the section with the smooth flow diminishes and morphs into jetting flow (the AoA is constant).

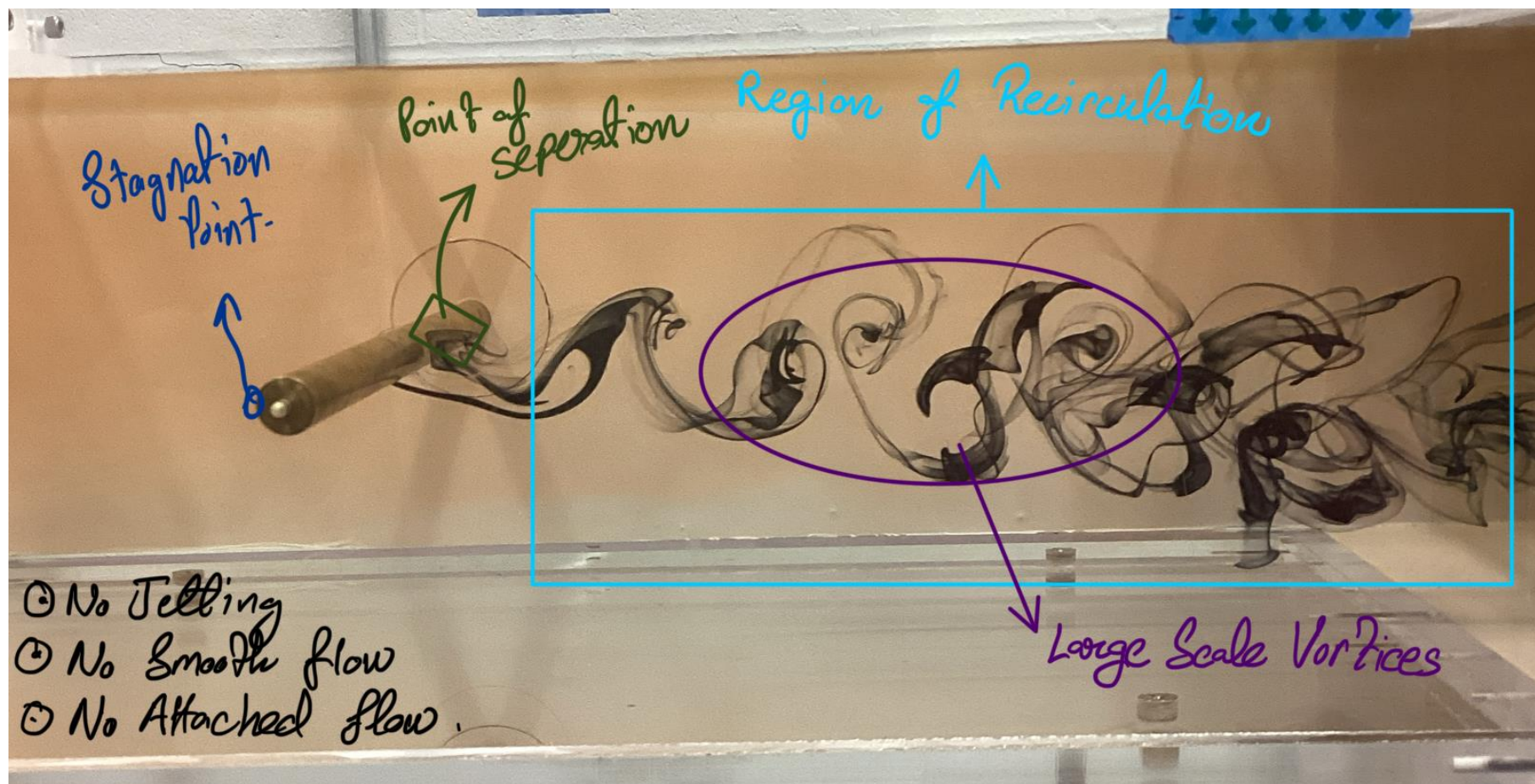
However, as the AoA increases to near stall angle, the flow is observed to detach/separate from the top section of the airfoil, followed by a region of recirculation and large vortices as the fluid flows past the trailing edge of the airfoil.

2. When the streamlines are attached to the airfoil and follow it closely, there is a pressure difference created as the fluid going under the airfoil has a higher pressure than the fluid going over it. This pressure difference is indicative of lift generation. However, at higher AoA (near stall) the flow separates from the airfoil which results in negligible pressure difference resulting in no lift. Furthermore, due to high AoA, the area of contact increases significantly and it results in a much higher drag when compared to low AoA & attached flow scenario.

CYLINDER RESULTS & DISCUSSION



Cylinder - Result



- Estimated Reynold's Number: **~200**
- Measured AoA: **N/A**

Cylinder - Discussion

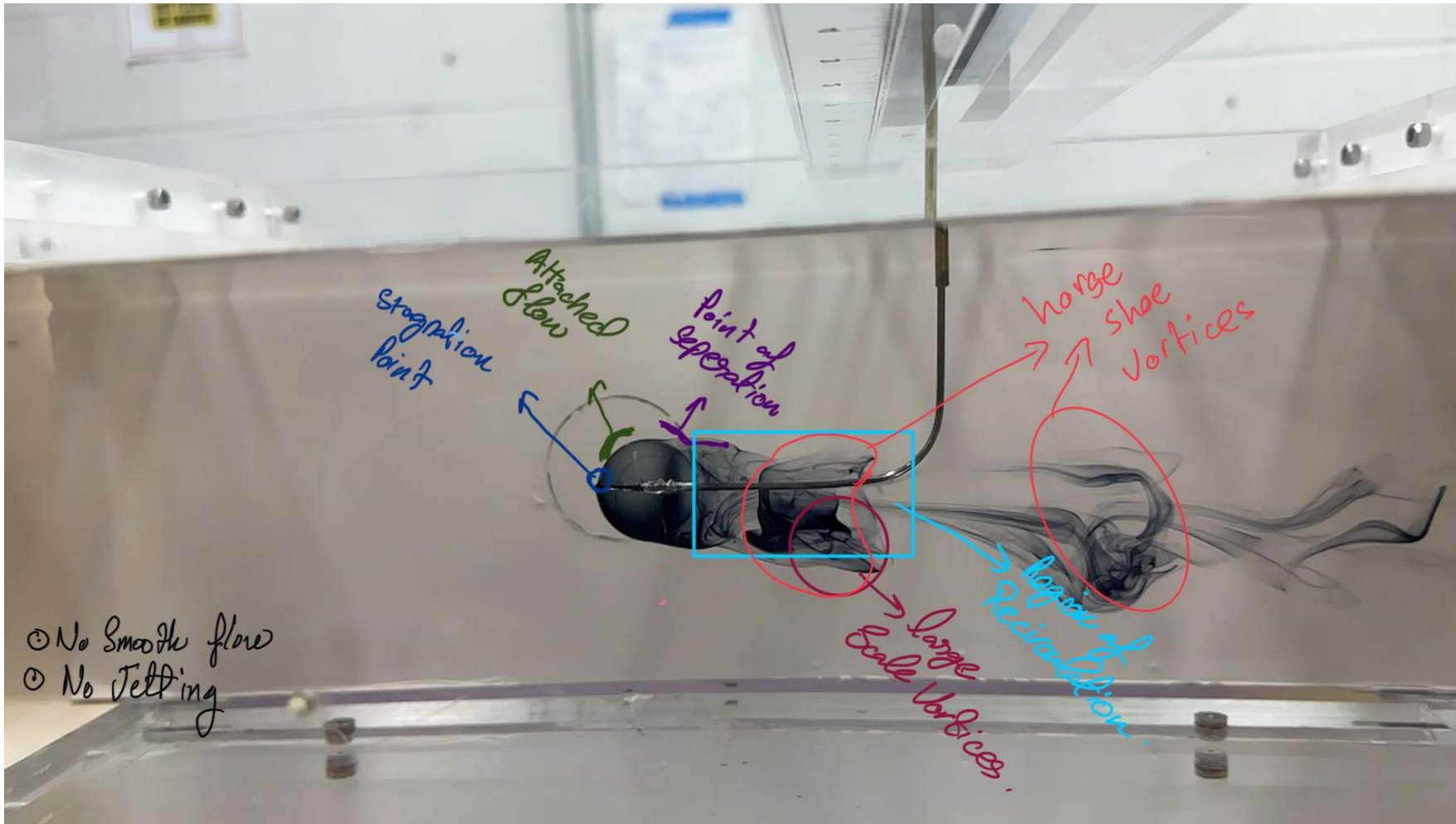
- The stagnation point of the flow for a cylinder is on the point facing the flow head on. By just observing the streamlines visually, there is a lot of turbulence as soon as the flow leaves the trailing edge of the cylinder. The flow immediately separates, and there are regions of large vortices and recirculation. (The vortices also seem to follow an oscillating zig-zag pattern)
- This indicates that there is a significant drag force generated from the cylinder in the fluid flow and also that there is negligible lift forces being generated.



SPHERE RESULTS & DISCUSSION



Sphere - Result



- Estimated Reynold's Number: **~500**
- Measured AoA: **N/A**

Sphere - Discussion

- Wake pattern of the vortices are indicative of high drag forces that are being experienced by the sphere in the flowing fluid. The drag force directed would be in the direction of the fluid flow.
- The wake pattern looks like it follows a straight path behind the sphere as the flow progresses from left to right.
- This wake pattern, as per my intuition, would also be a similar or an approximation of the path that the sphere might take as it falls towards Earth at terminal velocity.
- In conclusion, the sphere will fall in a straight path as it falls towards Earth at terminal velocity, given that it is not rotating (rotation will induce the Magnus Effect, which will make the path curve).

